

A global perspective on the probability of propagation of drought: from meteorological to soil moisture

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Abstract: With a copula-based probabilistic model, this study presents a global view on the propagation of meteorological to soil moisture drought. Meta-Gaussian and three-dimensional vine copulas were employed to construct the bivariate and trivariate conditional joint drought distributions, respectively, using the time series of standardized precipitation evapotranspiration index (SPEI) and standardized soil moisture index (SMI) derived from the Global Land Data Assimilation System (GLDAS) simulations. Three different cases, including the probability and magnitude of soil moisture drought (SMI) conditioned on 1-month meteorological drought (SPEI1), seasonal meteorological drought (SPEI3), and the joint effects of SPEI1 and antecedent soil moisture conditions (SMI_{lag-1}), were analyzed. Then, the duration of soil moisture drought in response to meteorological drought, and potentially affected cropland area were analyzed to assess potential agricultural impacts. Results showed that the likelihood of soil moisture drought conditioned on

23 SPEI1 in the Amazon rainforest, southern and eastern China, Southeast Asia, southeastern United
24 States, and west Africa was generally higher than in other regions. Under the impact of seasonal
25 meteorological drought, except for the Qinghai-Tibet Plateau in China, northern Russia, and North
26 Africa, the probability of soil moisture drought occurrence overall increased with significant
27 increments in Europe and North America. The antecedent soil moisture condition also influenced the
28 probability of drought propagation, and this effect was particularly significant for Northern China,
29 Russia, the U.S. Midwest, Canada, and Australia. Spatially, the western U.S., northern Canada, South
30 Africa, eastern Europe, North China, northern Russia, and most parts of Australia present longer
31 duration of soil moisture drought in response to meteorological drought. Comprehensive
32 considerations of the conditional probability, duration, and potentially affected area of soil moisture
33 drought for croplands revealed that among five continents, South America was more vulnerable to
34 agricultural drought under the condition of meteorological drought.

35 **Keywords:** Drought propagation; Probability; Meteorological drought; Soil moisture drought

1 Introduction

Drought is a recurring natural phenomenon that occurs in virtually every region of the world (Wilhite and Glantz, 1985). In the context of global warming, recent years witnessed the growing frequency of extreme drought events (e.g., the Millennium drought in Australia; the 2010 droughts in Amazon, southwestern China, and Russia; the 2011 droughts in Texas-Mexico and East Africa; and the U.S. summer droughts in 2012 and 2016), which caused devastating impacts on agriculture, economy, and human society (Lewis et al., 2011; Schubert et al., 2014; Hao et al. 2014; Williams et al., 2015; Otkin et al. 2018; Liu et al. 2020). To better cope with drought and alleviate its impacts, it is essential to understand how droughts evolve and proceed at the atmosphere-land interface.

Using historical observations and model projections, a multitude of studies have investigated the drying or wetting patterns at regional and global scales. The drought conditions under climate change have been established using the long-term series of various drought indices and their trends (e.g., Sheffield et al. 2012; Dai, 2013; Wang et al. 2015; Zhu et al. 2018). Several studies extracted historical droughts and analyzed their magnitude or characteristics (duration, intensity, affected area, and severity) and spatiotemporal migration patterns, using the threshold method (Sheffield and Wood, 2007; Herrera-Estrada et al. 2017; Zhu et al. 2019; Gu et al. 2020). Focusing on the uncertainties in the assessment of drought, some studies discussed the impacts of data selection (e.g., length and temporal coverage of data set), formulation of hydrological components (e.g., evapotranspiration), and drought indices (Trenberth et al. 2014). These studies provided valuable insights into trends and characteristics of drought; however, the links between different drought types were neglected in their evaluation.

From the perspective of causative mechanisms, the evolution of drought involves a complicated

58 transition of moisture deficiencies. The anomalous meteorological conditions resulting in
59 precipitation deficit and expanding evaporative demand can travel through the terrestrial part of the
60 hydrological cycle, leading to the depletion of soil moisture and ultimately developing into a
61 hydrological drought (Van Loon, 2015; Liu et al. 2017; Han et al., 2019; Hellwig et al., 2020; Wu et
62 al. 2020). The transition of drought signal from one type of drought to another is referred to as
63 drought propagation (Haslinger; 2014; Apurv et al., 2017). This suggests that a certain region may
64 suffer more than one drought type at the same time. Although several studies investigated the nexus
65 between different drought types and drought propagation features (e.g., Van Loon and Van Lanen
66 2012; Wong et al. 2013; Yang et al. 2017; Liu et al. 2019), these were mostly case studies and a
67 global evaluation of drought status with the consideration of propagation behavior was rare.

68 In this study, we focused on soil moisture drought as the direct successor of meteorological
69 drought, and used a probabilistic method to depict the process of drought propagation from a global
70 perspective. Two issues were addressed: One referred to the probability of soil moisture drought
71 conditioned on meteorological drought (to recognize globally sensitive regions vulnerable to
72 meteorological drought), and the other referred to the characteristics of soil moisture droughts based
73 on the established conditional joint distributions, including the probable magnitude of soil moisture
74 drought (i.e., the extent to which the soil moisture condition may be drying) given meteorological
75 drought at different severity levels, the extended duration of soil moisture droughts in response to
76 meteorological droughts, and potential agricultural drought risk analysis. The reminder of this paper
77 is organized as follows: Section 2 provides a brief description of the study area and data used in this
78 study. Section 3 describes the probabilistic model for inferring the probability of soil moisture
79 drought and the corresponding drought characteristics given meteorological drought under various

80 scenarios. Section 4 presents the results, Section 5 discusses the findings, and conclusions are drawn
81 in Section 6.

82 **2 Data**

83 **2.1 Global Land Data Assimilation System**

84 The Global Land Data Assimilation System (GLDAS) is designed to provide globally estimated
85 land surface fluxes and storages of water and energy (e.g., actual evapotranspiration, soil moisture,
86 surface runoff, and subsurface runoff) with finer spatiotemporal resolutions (Rodell et al., 2004). The
87 version 1 of GLDAS (hereafter GLDAS-1) includes four land surface models, i.e., the Mosaic, Noah,
88 CLM, and VIC. Evaluation of the GLDAS-1 showed that it introduced unnatural trends and led to
89 highly uncertain forcing fields in 1995-1997 as a result of the switched forcing data sources
90 (Beaudoin et al., 2016). Aimed at creating more climatologically consistent data sets, recently, the
91 Version 2 of GLDAS (hereafter GLDAS-2) was released with the enhancement of updated model
92 version (only the simulations of the Noah model are available) and land surface parameters.

93 The GLDAS-2 consists of 3-hour, daily, and monthly (generated through temporal averaging or
94 summation of the 3-hour products) products at 0.25° and 1.0° spatial resolutions. It has two different
95 forcing data-derived components: One refers to the GLDAS-2.0, which was run entirely with the
96 Princeton meteorological forcing data, and the other was forced with a combination of model and
97 observation based forcing data sets (hereafter, GLDAS-2.1). In this study, the monthly product
98 (including precipitation, potential evapotranspiration (PET), and soil moisture) of GLDAS-2.0
99 (GLDAS_NOAH10_M_2.0) was used for drought analysis. This product was generated through the
100 temporal averaging of reprocessed 3-hour data, and is currently available from January 1948 to
101 December 2015.

2.2 International Soil Moisture Network

The International Soil Moisture Network (ISMN; <https://ismn.geo.tuwien.ac.at/en/>) is an integrated system which hosts globally quality-controlled and harmonized in-situ soil moisture measurements from various ground validation campaigns and operational networks (Dorigo et al., 2011). Since its establishment, the ISMN has experienced a rapid growth with increasing number of participating networks and stations. At present, ISMN contains data of 59 operational networks and 8398 field observation stations. The database stored includes soil water contents of different soil layers (in units of volumetric soil moisture ($\text{m}^3 \cdot \text{m}^{-3}$)), and relevant hydrometeorological variables (e.g., precipitation, and air and soil temperatures), and serves as an important resource for validating satellite retrievals and land surface model-based soil moisture products. In this study, the ISMN was used to validate the soil moisture datasets from GLDAS. Although the time period of the entire products in ISMN spanned from 1952 until the present, the long-term time series of in-situ soil moisture observations on a global basis are still insufficient. For instance, rather limited data are available in Africa and South America. Daily soil moisture data from ISMN were aggregated into monthly averages, and 2455 sites with the temporal data coverage exceeding 120 months were selected to validate the accuracy of GLDAS (see Figure 1 for spatial distribution).

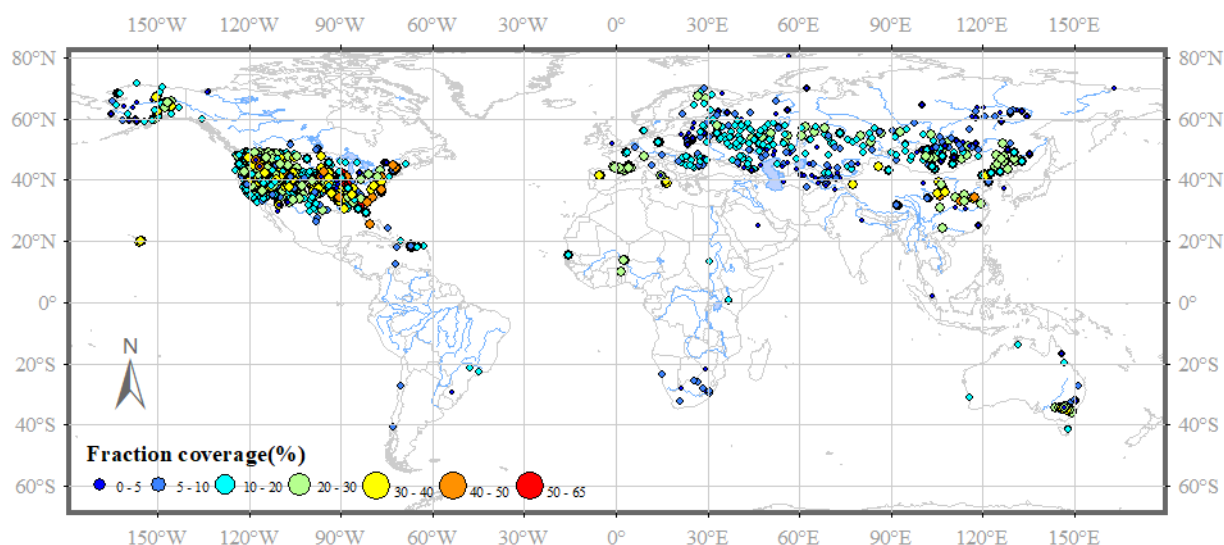


Figure 1. Map of the distribution of networks and stations contained in the ISMN. Color circles show the temporal coverage fraction for each network and station.

2.3 ESA CCI Land cover dataset

The global land cover product released by the European Space Agency (ESA) Climate Change Initiative (CCI) (<https://www.esa-landcover-cci.org/>) is employed for investigating the ground information. With the objective of providing a consistent historical land cover dataset for climate modeling related issues, this product is developed by merging multiple earth observation products of the ESA. The ESA CCI land cover dataset contains 24 consistent global land cover maps on an annual basis from 1992 to 2015, and the spatial resolution is 300 m. For defining land cover classes, the United Nations Land Cover Classification System which describes the terrestrial land surface in 37 original classes is adopted (Defourny et al. 2017). With strict quality control and validation, the accuracy of the ESA CCI land cover product is approximately 71.1%, with particularly high values (ranging between 83% and 92%) for cropland-related classes (Defourny et al. 2017). In this study, the most recent map series, namely the map of 2015 (v2.0.7), was used for analysis. Given a close relationship between cropland and soil moisture drought, we mainly focused on three land cover

types, i.e., rainfed cropland, irrigated cropland, and a mixture of cropland and natural vegetation.

3 Methods

3.1 Standardized drought index

The standardized precipitation evapotranspiration index (SPEI) (Vicente-Serrano et al., 2010) and standardized soil moisture index (SMI) (Sheffield et al., 2004) were employed to decipher meteorological and soil moisture droughts, respectively. These two drought indices follow the mathematical algorithm of the standardized precipitation index (SPI) (McKee et al., 1993), which uses the normal quantile transformation to standardize the index (namely to make the drought index spatiotemporally comparable). The difference between the two indices mainly lies in the variables incorporated for estimating the moisture status. For SPEI, the difference between precipitation and PET was taken as input, while for SMI, only the soil moisture content was considered. Like SPI, SPEI and SMI can also be calculated at multiple time-scales by accumulating the moisture deficits in any predetermined time period (e.g., 1–24 months). The SMI and SPEI share the same classification criteria: values of the index less than -0.5 correspond to mild drought, less than -1 to moderate drought, less than -1.5 to severe drought, and less than -2 to extreme drought. In this study, SMI was accumulated at a 1-month time scale, and SPEI was accumulated at 1-month and 3-month time scales, respectively (denoted as SPEI1 and SPEI3), to investigate the influence of meteorological moisture deficits in the current month and its cumulative effect on the soil moisture drought.

3.2 Probabilistic drought analysis using copulas

Copulas are widely used for constructing the joint distributions due to their flexibility to model the dependence structure among random variables regardless of their marginals. According to Sklar's theorem, for n -dimensional continuous random variables $(x_1, \dots, x_n)$ with marginal cumulative

156 distributions $F_1(x_1), \dots, F_n(x_n)$, there always exists an n -copula which can combine these univariate
 157 marginal distributions into a joint distribution function (Nelsen, 2007):

$$158 \quad H(x_1, \dots, x_n) = C(F_1(x_1), \dots, F_n(x_n)) = C(u_1, \dots, u_n) \quad (1)$$

159 where $H(\cdot)$ represents the joint distribution of random variables; C is the copula function satisfying
 160 $[0,1]^n \rightarrow [0,1]$; $F_i(x_i)$, denoted by u_i ($i=1, \dots, n$) in the copula function, represents the marginal
 161 distribution. The joint density function $f(x_1, \dots, x_n)$ can be decomposed using the copula density
 162 function as:

$$163 \quad f(x_1, \dots, x_n) = \left[\prod_{i=1}^n f(x_i) \right] \times c(u_1, \dots, u_n) \quad (2)$$

164 where $f_i(x_i)$ and c represent the marginal probability density of the marginal function and the copula
 165 function, respectively. Based on the copula concept, the conditional distribution of these variables
 166 can be derived to realize probabilistic drought prediction by establishing the nonlinear dependence
 167 between the predictand and predictors. In this study, we used variables (X_1, X_2, X_3) to represent
 168 meteorological drought (SPEI), antecedent soil moisture condition (denoted as $\text{SMI}_{\text{lag-1}}$), and soil
 169 moisture drought (SMI), respectively. By constructing their joint distributions, we investigated the
 170 probability of soil moisture drought under different scenarios:

- 171 (1) The probability of soil moisture drought (X_3) conditioned on meteorological drought (X_1).
- 172 (2) The probability of soil moisture drought (X_3) conditioned on meteorological drought (X_1) and
 173 antecedent soil moisture condition (X_2).

174 Figure 2 presents the framework of copula-based probabilistic drought analysis. Obviously, the first
 175 scenario belongs to a bivariate case, and the meta-Gaussian copula was employed to construct the
 176 joint distribution. For the second scenario, vine copulas were selected to establish the trivariate joint
 177 distribution. Detailed formulas for the above two types of copulas are given below.

3.2.1 Meta-Gaussian Copula

Given the capability of the Gaussian copula for modeling both positive and negative dependences among random variables, the bivariate form of the Gaussian copula was employed to construct the joint distribution of (X_I, X_3) :

$$C(u_1, u_3) = \Phi_2(\Phi^{-1}(u_1), \Phi^{-1}(u_3)) \\ = \int_{-\infty}^{\Phi^{-1}(u_1)} \int_{-\infty}^{\Phi^{-1}(u_3)} \frac{1}{(2\pi)\sqrt{1-\rho^2}} \exp\left\{-\frac{t^2 - 2\rho st + s^2}{2(1-\rho^2)}\right\} ds dt \quad (3)$$

where Φ is the cumulative distribution function of the standard normal distribution function, and Φ^{-1} represents the corresponding inverse function; ρ is the dependence parameter for the Gaussian copula; s and t are integral variables. Accordingly, the conditional distribution of X_3 given $X_I = x_I'$ (or u_I given u_I') can be expressed as:

$$P(X_3 \leq x_3 | X_I = x_I') = \frac{\partial C(u_1, u_3)}{\partial u_1} \Big|_{u = u'} \quad (4)$$

For the meta-Gaussian copula, the conditional distribution in equation (4) can be expressed as (Aas et al., 2009):

$$P(X_3 \leq x_3 | X_I = x_I') = \Phi\left(\frac{\Phi^{-1}(u_3) - \rho\Phi^{-1}(u')}{\sqrt{1-\rho^2}}\right) \quad (5)$$

Since both X_I and X_3 are the standardized drought indices (i.e., normally distributed), it is generally reasonable to assume a multivariate normal distribution function to model their joint behavior (Wilks, 2011; Hao et al., 2016). Accordingly, for normal random variables X_I and X_3 with the joint normal distribution, the analytical form of the conditional distribution of X_3 given X_I is given as:

$$X_3 | X_I \sim N(\mu_{X_3|X_I}, \Sigma_{X_3|X_I}) \quad (6)$$

where $\mu_{X_3|X_I}$ and $\Sigma_{X_3|X_I}$ represent the conditional mean (i.e., the peak of the conditional probability density distribution curve in Figure 2, and is referred to as the most probable predicted value) and

conditional covariance matrix, respectively, which can be derived as:

$$\boldsymbol{\mu}_{X_3|X_1} = \boldsymbol{\mu}_{x_3} + \boldsymbol{\Sigma}_{x_3x_1} \boldsymbol{\Sigma}_{x_1x_1}^{-1} (x - \boldsymbol{\mu}_{x_1}) \quad (7)$$

$$\boldsymbol{\Sigma}_{X_3|X_1} = \boldsymbol{\Sigma}_{x_3x_3} - \boldsymbol{\Sigma}_{x_3x_1} \boldsymbol{\Sigma}_{x_1x_1}^{-1} \boldsymbol{\Sigma}_{x_1x_3} \quad (8)$$

where $\boldsymbol{\mu}_{x_1}$ and $\boldsymbol{\mu}_{x_3}$ are means of X_1 and X_3 , and $\boldsymbol{\Sigma}_{x_1x_1}$, $\boldsymbol{\Sigma}_{x_1x_3}$, $\boldsymbol{\Sigma}_{x_3x_1}$, and $\boldsymbol{\Sigma}_{x_3x_3}$ are the covariance matrix.

Detailed formulas for calculating the covariance matrix can be found in Hao et al. (2016, 2017).

3.2.2 Vine Copula

Vine copulas are effective tools for modeling flexible dependences in high dimensions without requiring a conditional independence assumption (Aas et al., 2009; Liu et al., 2016; 2018). Due to its flexibility in dependence modeling, the vine copula shows its superiority for constructing the conditional dependence, asymmetries, and tail dependence. These copulas are based on pair copula constructions (PCCs) which decompose an n -dimensional multivariate density into $n(n-1)/2$ bivariate copula densities (Aas et al., 2009). For a vine copula structure, $n(n-1)/2$ pair copulas are arranged in $n-1$ trees (Kurowicka and Cooke, 2006; Brechmann et al., 2013). There are two general types of vine copulas, i.e., canonical vines (C-vines) and drawable vines (D-vines). In this study, we used the C-vines to construct the joint distribution. The n -dimensional density corresponding to a canonical vine is given by (Aas et al., 2009):

$$f(x_1, \dots, x_n) = \prod_{i=1}^n f(x_i) \times \prod_{i=1}^{n-1} \prod_{j=1}^{n-i} c_{i,i+j|1:(i-1)}(F(x_i | x_1, \dots, x_{i-1}), F(x_{i+j} | x_1, \dots, x_{i-1})) \quad (9)$$

where $f(x_1, \dots, x_n)$ denotes the joint density function, $f(x_i)$ are the marginal densities, and $c_{i,i+j|1:(i-1)}$ represents the bivariate copula densities. To construct the C-vine copula, five bivariate copulas (i.e., Student t , Gaussian, Clayton, Frank, and Gumbel-Hougaard) were selected as candidate pair copulas, and the optimal one for each pair copula was chosen through the Akaike information criterion

219 (Schepsmeier and Brechmann, 2015). For the trivariate case, the three-dimensional density function
 220 can be decomposed into three bivariate copula densities and their margins:

$$221 \quad f_{123} = f_1 \cdot f_2 \cdot f_3 \cdot c_{12} \cdot c_{13} \cdot c_{23|1} \quad (10)$$

222 where c_{12} is the abbreviated form of $c_{1,2}(F(x_1), F(x_2))$.
 223

224 The following step was used to to compute the conditional distribution functions and conditional
 225 bivariate copulas in equation (10). According to Joe (1996) and Aas et al. (2009), the conditional
 226 distribution function $F(x|w)$ for an m -dimensional vector $w = (w_1, \dots, w_m)$ can be obtained through the
 following recursive function:

$$227 \quad h(x|w) := F(x|w) = \frac{\partial C_{xw_j|w_{-j}}(F(x|w_{-j}), F(w_j|w_{-j}))}{\partial F(w_j|w_{-j})} \quad (11)$$

228 where w_j ($j=1, \dots, m$) denotes any element in w , w_{-j} represents the vector without element w_j , and
 229 $C_{xw_j|w_{-j}}$ is the bivariate copula function. Let $u_i (i=1, \dots, n) = F(x_i)$. Then, the trivariate case of the
 230 conditional distribution function in equation (11) can be written as:

$$231 \quad F(u_3|u_1, u_2) = \frac{\partial C_{u_3, u_2|u_1}(F(u_3|u_1), F(u_2|u_1))}{\partial F(u_2|u_1)} \quad (12)$$

232 Let

$$233 \quad F(u_3|u_1) = h(u_3|u_1) = \frac{\partial C_{u_3, u_1}(F(u_3), F(u_1))}{\partial F(u_1)}$$

234 and

$$235 \quad F(u_2|u_1) = h(u_2|u_1) = \frac{\partial C_{u_2, u_1}(F(u_2), F(u_1))}{\partial F(u_1)}$$

236 Accordingly, equation (12) can be written as:

$$237 \quad F(u_3|u_1, u_2) = h[h(u_3|u_1)|h(u_2|u_1)] \quad (13)$$

238 The inverse form of equation (13) can be applied for conditional simulations. Taking the bivariate
 239 case as an example, assuming $h(u_2|u_1)$ is the conditional distribution function of two random

variables x_1 and x_2 . For fixed probabilities τ (e.g., $\tau = 0.05, 0.1, \dots, 0.95$), u_2 can be estimated through an explicit function $u_2 = C_{u_2|u_1}^{-1}(\tau; u_1) = h^{-1}(\tau|u_1)$, where $C_{u_2|u_1}^{-1}$ is the inverse of τ quantile curve of the copula (Chen et al., 2009; Xu and Childs, 2013; Liu et al., 2015). The τ th copula-based conditional quantile function of variable x_2 can be written as:

$$x_2(\tau|x_1) = F^{-1}(C_{u_2|u_1}^{-1}(\tau; u_1)) = F^{-1}(h^{-1}(\tau|u_1)) \quad (14)$$

where F^{-1} is the inverse of u_2 .

Similarly, for the trivariate case, the variable x_3 given variables x_1 and x_2 can be derived as:

$$x_3(\tau|x_1, x_2) = F^{-1}\left\{h^{-1}\left[h^{-1}(\tau|h(u_2|u_1, \theta_{12}), \theta_{23|1})\right]u_1, \theta_{13}\right\} \quad (15)$$

where θ_{12} is the parameter of joint distribution c_{12} which measures the dependency of underlying conditional variables (x_1, x_2) . Likewise, θ_{13} and $\theta_{23|1}$ represent the parameters of c_{13} and $c_{23|1}$, respectively.

To obtain the best estimation of x_3 , the Monte Carlo simulations were employed to generate 1000 random numbers (i.e., τ) over the interval $[0, 1]$. Then, equation (15) was used to compute 1000 estimated values of x_3 , and the mean value was considered as the best estimate of x_3 .

3.3 Framework

The framework of copula-based drought propagation analysis is presented in Figure 2. Employing the time series of SPEI (1-month and 3-month time scales of SPEI were employed) and SMI (1-month time scale) generated from the GLDAS simulations and ISMN soil moisture observations, the bivariate and trivariate conditional joint distributions were constructed with meta-Gaussian copula and C-vine copula, respectively. Taking the case of SMI and SPEI1 as an example, the values of $P(\text{SMI}|\text{SPEI1})$ derived from GLDAS simulations were evaluated against those of ISMN in-situ measurements. Then, from a global perspective, the probability and most probable

magnitude of soil moisture drought (SMI) under varying scenarios were analyzed. These included the bivariate cases modeled by using the meta-Gaussian copula, i.e., the probability of soil moisture drought conditioned on 1-month meteorological drought (SPEI1), 3-month meteorological drought (SPEI3), and the triavariate case derived from vine copulas, i.e., the joint effects of SPEI1 and the antecedent soil moisture status (SMI_{lag-1}). Finally, based on the ESA CCI land cover map, a further investigation into the proportion of affected cropland area given meteorological drought was carried out to analyze the potential agricultural drought impacts.

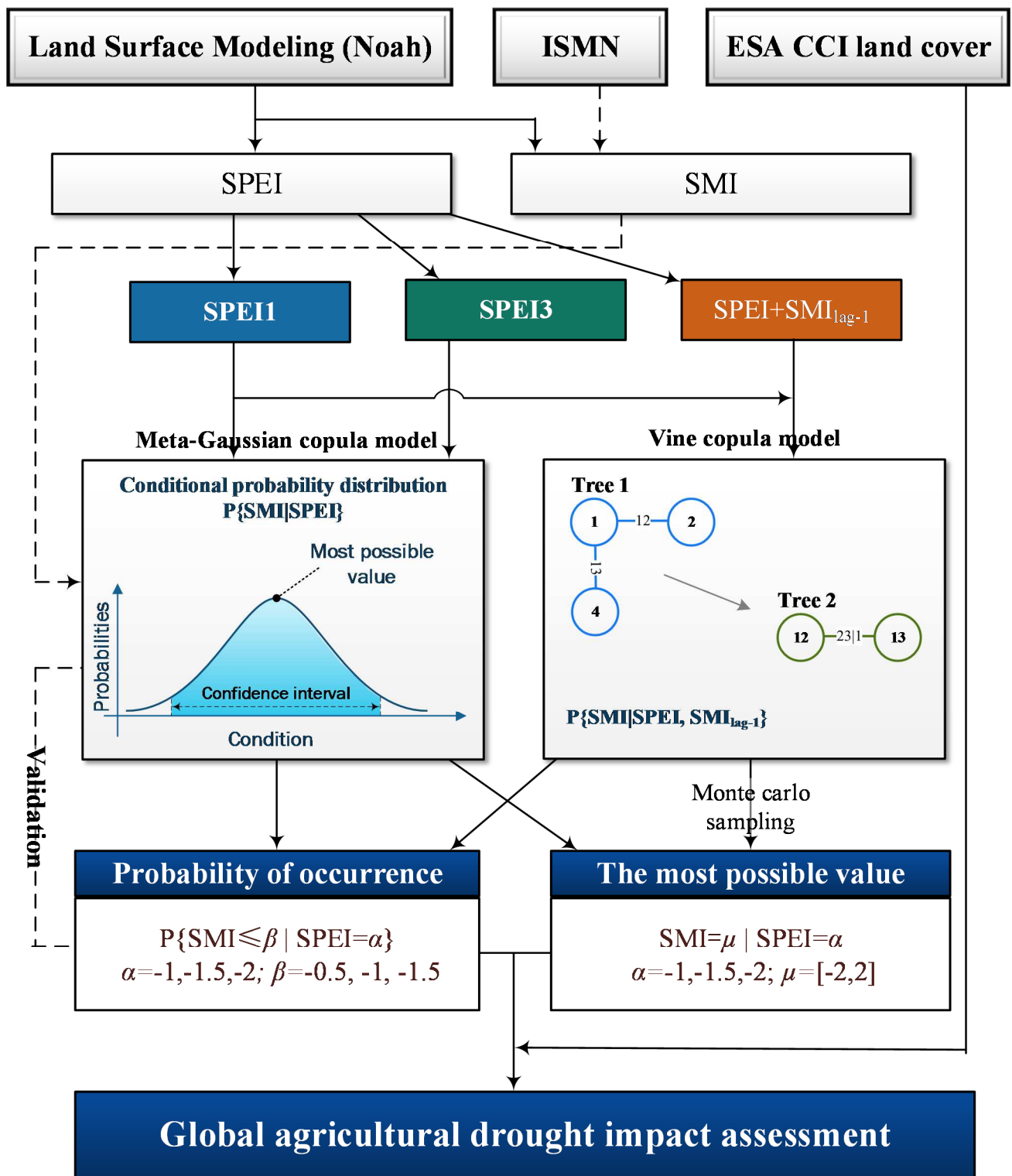


Figure 2. Framework of copula-based analysis for estimating the probability and most probable magnitude of soil moisture drought given meteorological drought under varying scenarios.

4 Results

4.1 GLDAS and in-situ comparison

To provide a direct assessment on the performance of GLDAS in estimating the probability of drought propagation, the conditional probabilities derived from GLDAS simulations were evaluated against those of ISMN stations. As an example, Figure 3 exhibits the relationship between soil moisture and meteorological droughts of different severity levels (i.e., moderate, severe, and extreme droughts when SPEI/SMI was no more than -1, -1.5, and -2, respectively) for the current month (i.e., SPEI1 was employed). The grey error bars in the figure represent the condition when several ISMN stations are located in the same grid cell of GLDAS, and the probabilities derived from GLDAS would correspond to a couple of ISMN-based values. As expected, some disparities were observed between the two estimated probability series with certain scatters from the 1:1 line. The scale mismatch between grid-based GLDAS data and point-based ISMN could be one reason. In particular, for areas where soil moisture exhibited large spatial heterogeneity, the estimated probability values could vary significantly from one site to another (these sites were located in the same grid cell of GLDAS), such as the large error bars in Figure 3. In addition, since most series of ISMN were intermittent, the different temporal coverages of soil moisture data employed for constructing the propagation relationship might also contribute to the discrepancies. Aside from the above mentioned aspects, there was generally a good agreement between the probabilities estimated from GLDAS and ISMN with the majority of scatters distributed around the 1:1 line, implying that the spatial pattern for the probability of drought propagation revealed by these two data sets were basically comparable.

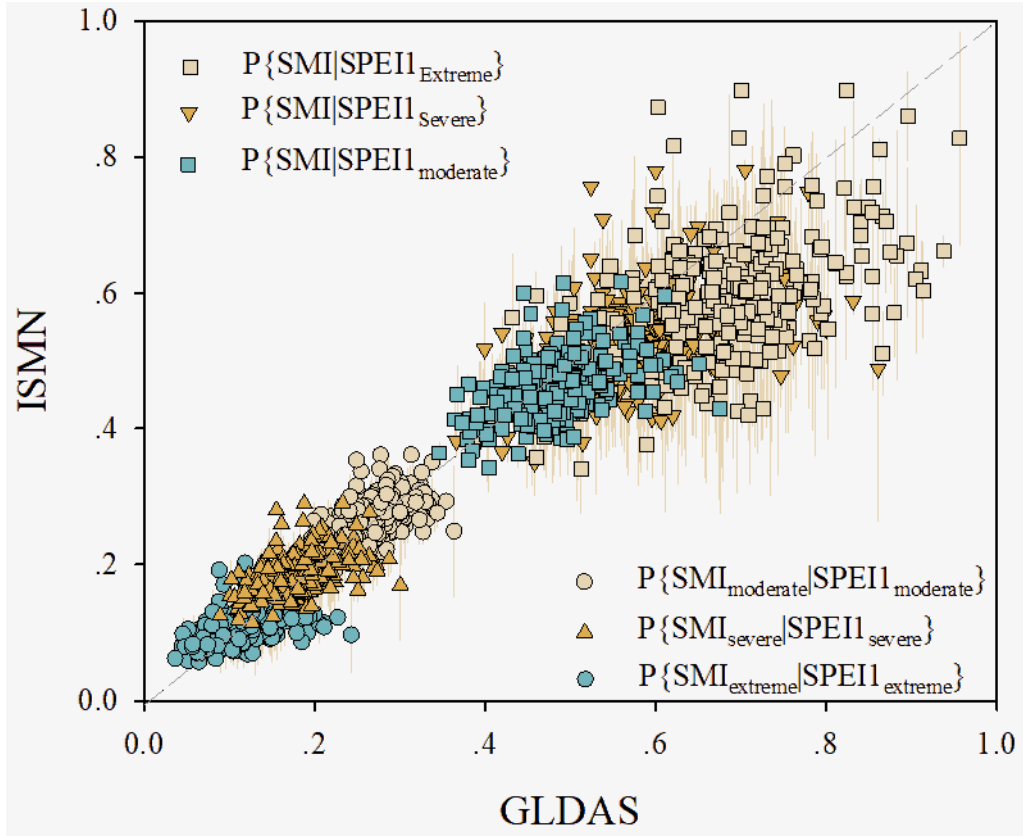


Figure 3. Comparison of the probability of soil moisture drought derived from GLDAS simulations against those of ISMN in-situ measurements. $P\{SMI|SPEI_{moderate}\}$, $P\{SMI|SPEI_{severe}\}$, and $P\{SMI|SPEI_{extreme}\}$ denote the probability of soil moisture under dry conditions (i.e., $SMI < -0.5$) when experiencing a meteorological drought at the moderate, severe, and extreme category, respectively. $P\{SMI_{moderate}|SPEI_{moderate}\}$, $P\{SMI_{severe}|SPEI_{severe}\}$, and $P\{SMI_{extreme}|SPEI_{extreme}\}$ denote the likelihood of moderate, severe, and extreme soil moisture drought conditioned on meteorological droughts of the same levels of drought severity. The grey error bars represent the condition when several ISMN stations are located in the same grid cell of GLDAS, and the probabilities based on GLDAS would correspond to a couple of ISMN based values.

307 **4.2 Probability of soil moisture drought**

308 The probability of drought propagation from meteorological to soil moisture under three
309 different cases was analyzed. Based on SMI and SPEI1, the spatial patterns of drought propagation
310 from meteorological to soil moisture for the current month were first detected. Then, SPEI1 was
311 replaced with SPEI3 to analyze the effect of seasonal meteorological droughts on soil moisture
312 droughts. Depletion of soil moisture storage is related to the initial condition of soil layer, and in the
313 third case, the antecedent soil moisture status was also employed as a conditional variable for
314 constructing the joint distribution.

315 **4.2.1 Probability of soil moisture drought conditioned on meteorological drought**

316 Figure 4 shows the probability of soil moisture drought given meteorological drought for the
317 current month (SPEI1) under eight scenarios. Specifically, panels in the same row exhibit the varying
318 probabilities of soil moisture drought given gradually enhanced signal of meteorological drought (i.e.,
319 conditioned on moderate, severe, and extreme meteorological drought from left to right), while
320 panels in the same column present the varying probabilities of soil moisture drought at mild,
321 moderate, and severe levels respectively, given the same condition of meteorological drought. Taking
322 the case of the first row as an example (Figures 4a, c, and f), the probability of soil moisture drought
323 occurrence significantly increased conditioned on meteorological drought of severity levels from
324 moderate to extreme, implying that the likelihood of soil moisture drought tended to be higher when
325 the climatic condition was drier. In particular, the scenario in Figure 4f reflected a typical condition
326 where most regions of the world (except the extremely dry North Africa and extremely cold
327 northeastern areas in Russia) exhibited rather high values (with probabilities above 0.6) for the case
328 under on an extreme condition, implying that meteorological drought of an extreme level was very

likely to propagate into soil moisture drought at a large scale. With globally low values (around 0.1-0.3) for Figures 4b and e, they revealed another phenomenon that the likelihood of soil moisture drought at the same severity level as meteorological drought was generally low. Besides, different regional behaviors on the probability of drought propagation could also be observed. Spatially, the Amazon rainforest, southern and eastern China, Southeast Asia, southeastern United States, West and Central Africa, were more sensitive to meteorological drought with the probabilities of soil moisture drought two to three fold higher than were other regions (Figure 4g).

Figure 5 shows the probability of soil moisture drought conditioned on SPEI3. It can be seen that except for the cases of soil moisture drought at the same severity level as meteorological drought (Figures 5b, e), the probability of drought propagation under the seasonal meteorological drought overall increased compared with Figure 4. Spatially, Europe and North America were two major regions which exhibited significant changes. Taking the occurrence of soil moisture drought given a severe condition (Figure 5c) as an example, compared to the pattern in Figure 4c, the probability of soil moisture drought given severe meteorological drought in Europe largely increased with the values varying from 0.5 to 0.8. The change in North America was mainly manifested as the enlarged area from the southeastern United States to the entire North America region (Figures 4c and 5c) with higher probability values. To explore the underlying reasons for such a spatial pattern, the regionally averaged time series of SPEI and SMI in different climate regions were analyzed. It can be seen that for the Amazon rainforest which presents the highest value of conditional probability, the SMI is sensitive to the changes of climatic condition and varies synchronously with SPEI (Figure 6a). A general good agreement between SMI and SPEI were also observed in Europe, however, for minor meteorological droughts such as the 1999 drought event (right panel) in Figure 6b, the soil moisture

351 presents no response to the dry climate condition. Similarly, the cases for the eastern and western
352 U.S. (Figure 7) also suggests a good consistency between the time series of SPEI and SMI
353 contributes to high conditional probability. In contrast, the Qinghai-Tibet Plateau in China, northern
354 Russia, and North Africa were less affected by the seasonal cumulative water deficits, and
355 maintained rather low probability values (around 0.1-0.4) even under an extreme meteorological
356 drought (Figure 5f). This may be related to the specific climate and geographical conditions of these
357 regions. The Qinghai-Tibet Plateau in China and northern Russia are typical permafrost and
358 snow-covered regions, where snow or permafrost acts as a natural reservoir to store water when the
359 temperature is below zero, and to release water to ameliorate water shortage as a result of snowmelt.
360 This leads to the land surface (e.g., soil moisture, and runoff) may have a slower response to
361 meteorological drought than other regions (Van Loon and Van Lanen, 2012; Qi et al., 2020; Liu et al.,
362 2020). As for the North Africa, it is the driest area in the world with annual mean precipitation no
363 more than 50 mm. Under this tropical desert climate, the soil layer maintains a dry status all the year
364 around and is insensitive to the changes of climatic condition.

365

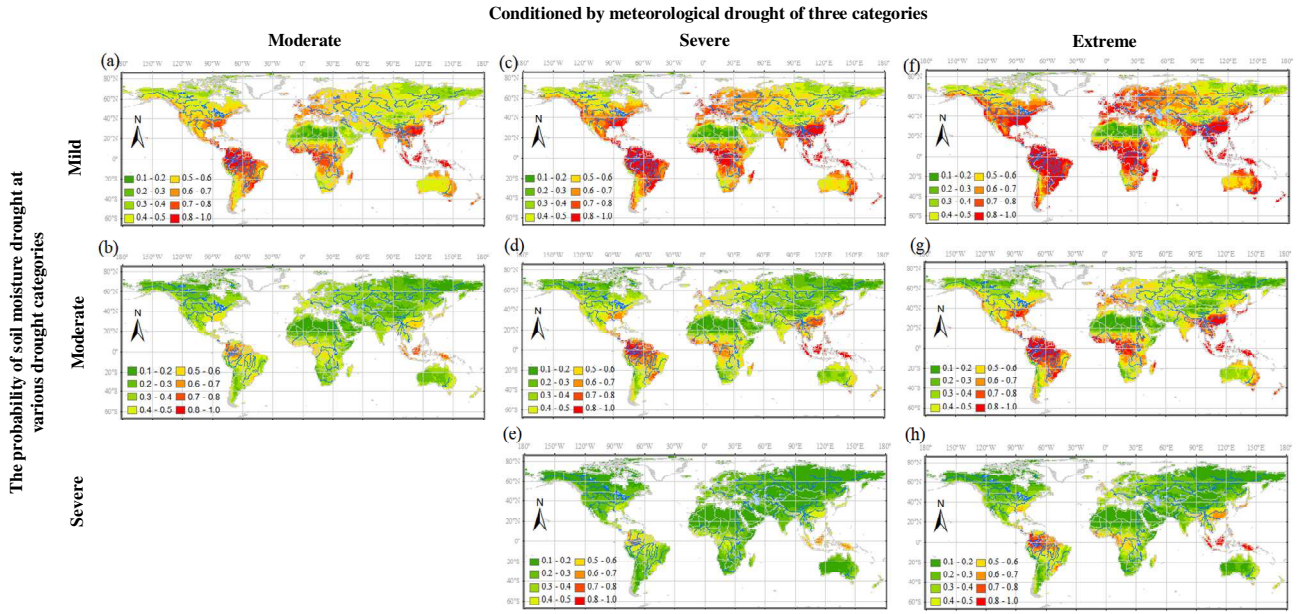


Figure 4. Probability of mild, moderate, and severe soil moisture drought under meteorological drought (indicated by SPEI1) of various severity levels (i.e., conditioned on moderate, severe, and extreme meteorological drought from the left column to the right column), respectively.

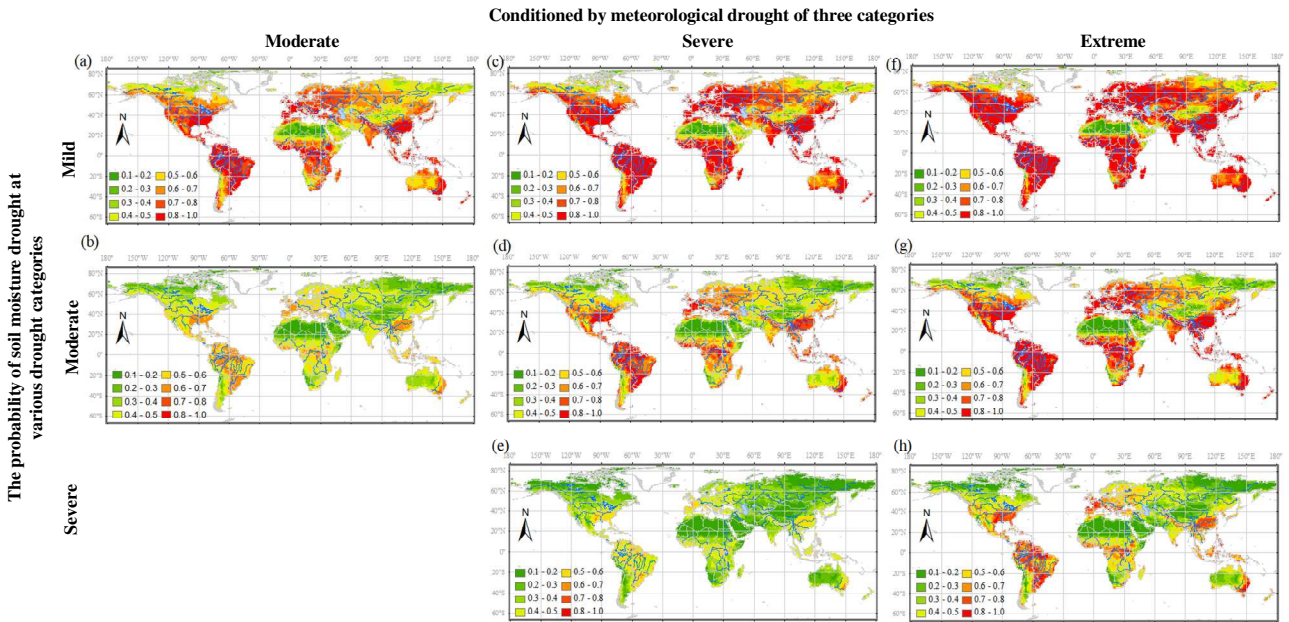


Figure 5. As in Figure 4 but conditioned on SPEI3.

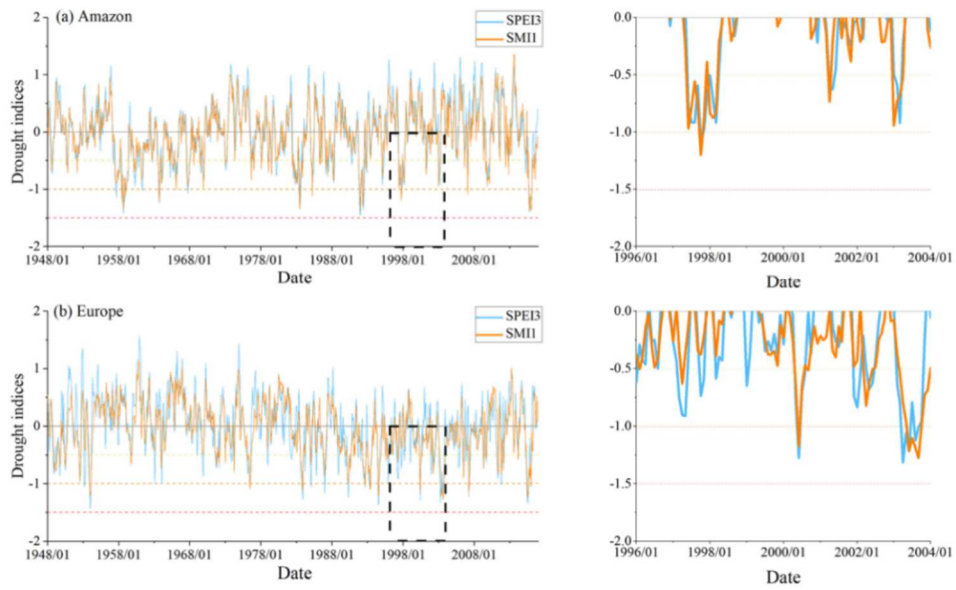


Figure 6. Regionally averaged time series of SPEI3 and SMI for (a) Amazon and (b) Europe. The right panels are the amplified time series of SPEI3 and SMI for dashed rectangle marked drought events in the left panels.

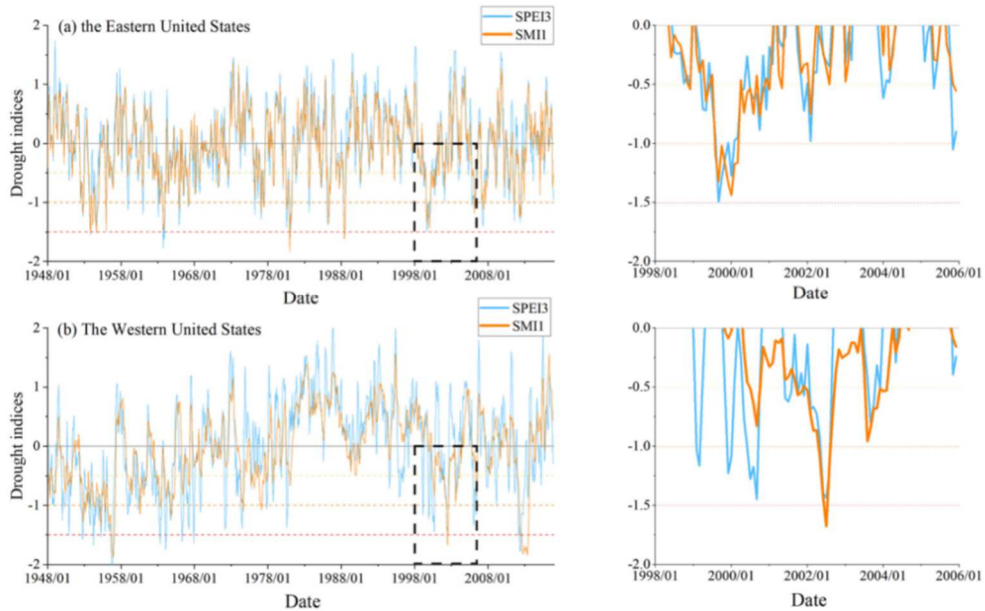
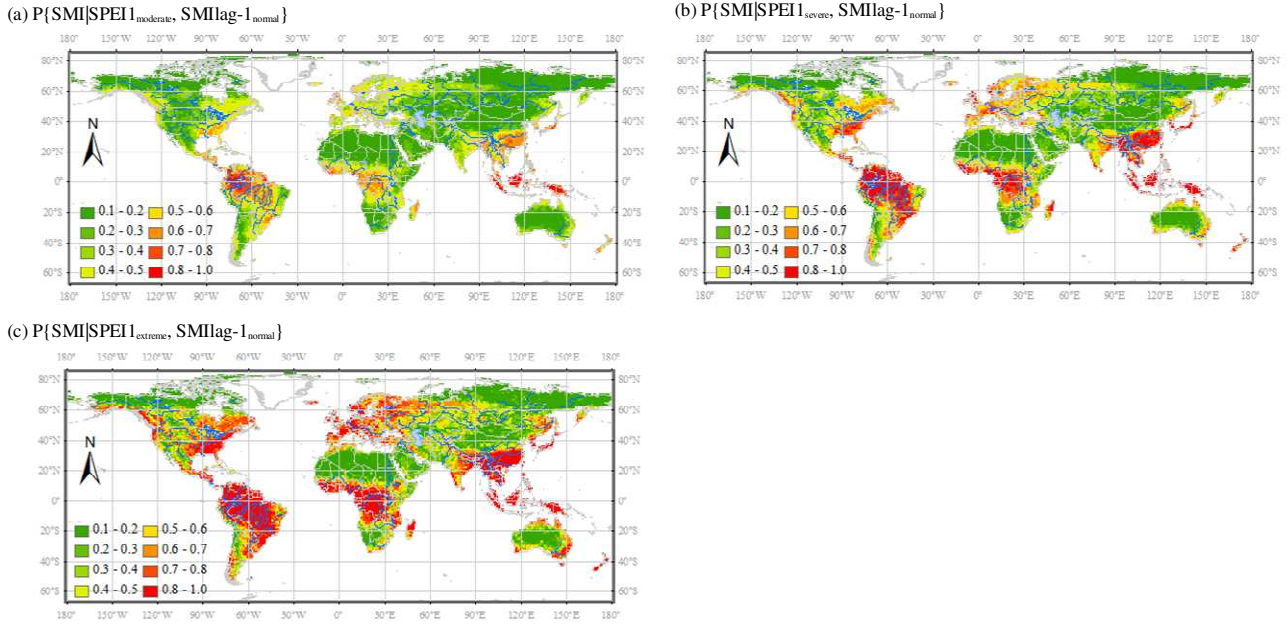


Figure 7. As in Figure 6 but for (a) eastern and (b) western U.S.

4.2.2 Probability of soil moisture drought conditioned on meteorological drought and antecedent soil moisture condition

Based on equations (10)-(13), the probability of soil moisture drought conditioned on meteorological drought and previous SMI values (denoted as SMI_{lag-1}) can be derived. Figure 8 shows the likelihood of soil moisture conditioned on meteorological droughts and normal antecedent soil moisture conditions ($-0.5 < SMI_{lag-1} < 0.5$). Comparing to the patterns in Figure 4, it can be seen that globally the probability values of soil moisture drought under the joint condition of moderate, severe, and extreme meteorological drought and normal antecedent soil moisture status generally decreased, with significant reduction (from 0.5/0.6 to 0.1/0.2) in Northern China, Russia, the American Midwest, Canada, and Australia. The significant changes for the conditional probability of soil moisture drought in these regions are possibly related to the soil moisture–temperature coupling effect. According to the geographical definition, these regions are mostly “transitional regions”, where soil moisture has been shown to have a strong influence on the land-atmospheric interactions via the soil moisture-temperature coupling effect (Seneviratne et al., 2010). This includes the feedback loops that a positive anomaly of temperature in response to a negative soil moisture anomaly, and increased temperature leads to a higher vapor pressure deficit and evaporative demand. Variations in these climate factors may contribute to a potential increase in evapotranspiration despite the dry conditions, which possibly leading to a further decrease in soil moisture. Obviously, soil moisture plays a key role for the drought propagation process from meteorological to soil moisture through its impact on land energy and water balances in transitional regions. Exceptional cases were observed for the Amazon rainforest, southern and eastern China, southeastern United States, west Africa, and Southeast Asia, where the probability values (above 0.7) were as high as in

405 Figure 4, indicating that the effect of antecedent soil moisture in these regions was generally minor.



406

407 Figure 8. Probability of soil moisture drought under the joint condition of (a) moderate, (b) severe,

408 and (c) extreme meteorological drought (indicated by SPEI1) and normal antecedent soil

409 moisture status (i.e., $-0.5 < SMI_{lag-1} < 0.5$).

410 4.3 Magnitude of soil moisture drought

411 Apart from the probability of soil moisture drought, it is also essential to figure out the most

412 probable magnitude of drought (e.g., moderate or severe drought), which has significance for

413 drought mitigation and drought early warning. Based on the conditional probability distribution, the

414 SMI values could be estimated through equations (7) and (15) for the bivariate (i.e., SMI conditioned

415 on SPEI) and trivariate cases (i.e., SMI conditioned on SPEI and SMI_{lag-1}), respectively. This may

416 produce SMI values of a wide range but with different probabilities (see the conditional probability

417 density distribution curve in Figure 2). For example, supposing the estimated range of SMI under a

418 severe meteorological drought (i.e., $-1.5 \leq SPEI < -2$) is -1.5 to 0.3 , this suggests the magnitude of soil

419 moisture drought might be one of the normal, mild, and moderate drought categories. From the

conditional probability density distribution curve, we can see that the likelihood becomes larger when the interval of estimated SMI values getting close to the mean. Obviously, the interval of estimated SMI values mostly close to mean is the most probable magnitude of a soil moisture drought, and is exactly the information we are concerned. Figure 9 exhibits the extent to which the soil moisture condition may be drying under six different scenarios. It can be seen that the estimated magnitude of soil moisture drought conditioned on SPEI1 was generally comparable to that conditioned on SPEI1 and SMI_{lag-1} (e.g., Figure 9a versus 9d). For most parts of the world, a moderate meteorological drought may cause little effect on the soil moisture condition, while severe and extreme meteorological drought may lead to mild soil moisture drought. However, for sensitive regions, such as the Amazon rainforest, southern and eastern China, west Africa, southeastern United States, and Southeast Asia, a mild soil moisture drought is likely to occur given moderate meteorological drought, and the soil moisture condition may become more drier given severe or extreme meteorological drought.

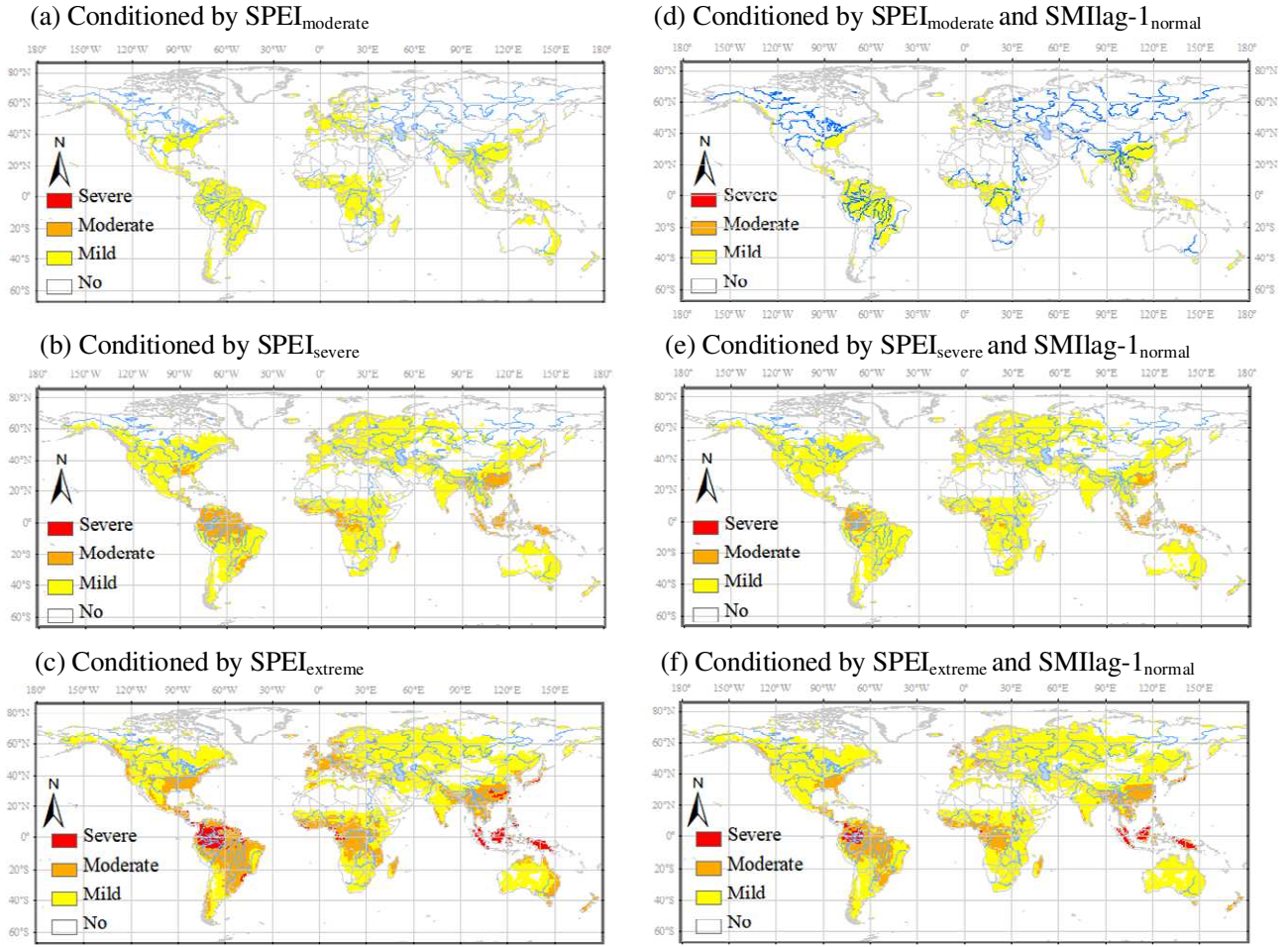


Figure 9. Magnitude of soil moisture drought under different scenarios. The left column shows the cases conditioned on SPEI1, while the right column shows the cases under the joint condition of SPEI1 and $SMlag-1_{\text{normal}}$.

4.4 Extended duration of soil moisture drought

The duration of soil moisture droughts provides important information for estimating potential drought impacts on crop productivity. To investigate how the meteorological droughts affect the duration of soil moisture drought, Figure 10 compares the average duration of soil moisture drought events ($SMI1$) against that of meteorological drought events ($SPEI1$). Positive values indicate soil moisture drought lasts longer than meteorological drought, and negative values indicate soil moisture drought lasts shorter than meteorological drought. As shown in Figure 10, for most regions of the

world, the duration of soil moisture drought is overall longer than that of meteorological drought. This phenomenon is generally consistent with findings of previous studies (e.g., Van Loon, 2015), which claimed that the duration of drought would be lengthened during the drought propagation process from atmosphere to the land surface. Given different climates and underlying surfaces, the terrestrial water storage (e.g., soil moisture, surface water, and groundwater) may present different responses with lengthened duration of varying degrees. Spatially, the western U.S., northern Canada, South Africa, eastern Europe, North China, northern Russia, and most parts of Australia present larger values, where the average duration of soil moisture drought events are more than three months longer than that of meteorological drought events. The time series of SPEI3 and SMI1 for selected regions in Figures 6 and 7 also revealed this difference in terms of drought duration. For instance, from Figures 6 and 7 it can also be seen that in regions with low conditional probabilities like Europe and western U.S. (climatically Europe is drier than Amazon, and western U.S. is drier than eastern U.S.), although the meteorological droughts do not immediately initiate soil moisture droughts, once the soil moisture droughts do get initiated, they last longer than meteorological droughts. This is consistent with findings in Sheffield et al. (2004) which also showed higher persistence of soil moisture droughts in western U.S. compared to eastern U.S. Several grids in the west and north Africa are the exception, where negative values were observed. This suggests the soil moisture drought events on average, persist shorter than meteorological drought events. Meanwhile, it should be noted that the uncertainties of GLDAS in west and north Africa may also be responsible for these negative values (see details in Section 5.1).

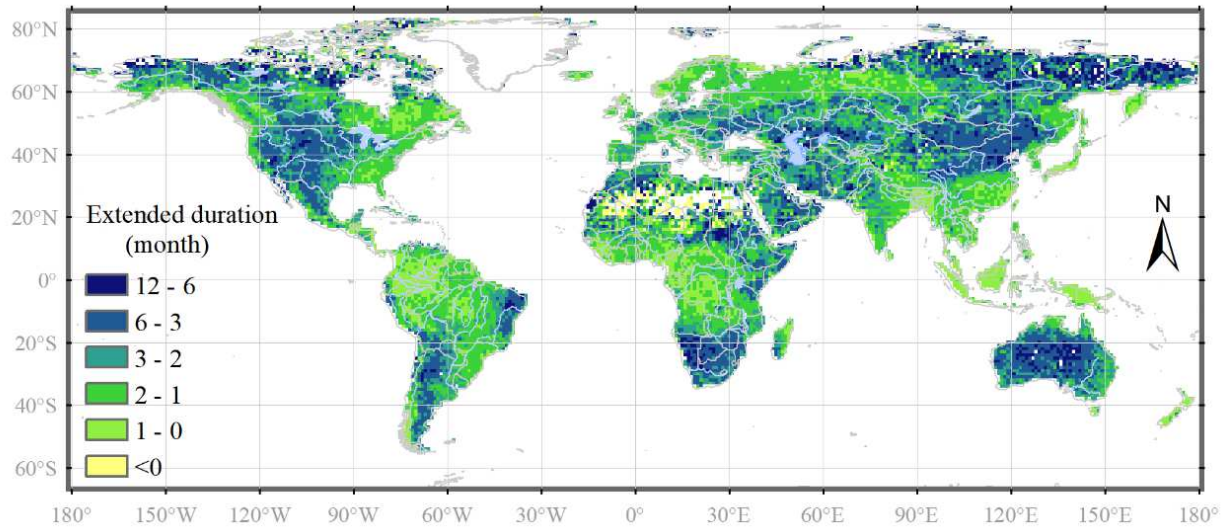


Figure 10. The average extended duration of soil moisture drought (SMI1) comparing to meteorological drought (SPEI1). Positive values indicate soil moisture drought lasts longer than meteorological drought, and negative values indicate soil moisture drought lasts shorter than meteorological drought.

4.5 Global agricultural drought impact analysis

Soil moisture has a close relationship with crop productivity, and one of the major impacts of soil moisture drought is agricultural losses (croplands) which threaten the food security. When a meteorological drought occurs, the likelihood of a soil moisture drought in different regions could be estimated through their conditional joint distributions. The occurrence of a soil moisture drought means reduced supply of moisture to the cropland, thus estimated conditional probability of the dry condition in soil layer may have some implications for recognizing the potential agricultural drought impacts. Besides, the duration of soil moisture drought (reflecting the persistence of moisture deficits of the soil layer), and its spatial coverage (reflecting how much area suffers from droughts) are also informative for estimating the strength of agricultural impacts. In this section, comprehensive considerations of drought propagation features, including the conditional probability, extended

480 duration, and affected area of soil moisture droughts in response to meteorological droughts for
481 croplands were provided to assess globally potential agricultural drought impacts.

482 Figure 11 presents the spatial distribution of croplands, and the proportion of affected crop area
483 by mild, moderate, severe, and extreme drought given meteorological drought (indicated by SPEI1)
484 for globe and each continent. As shown in Figure 11a, according to the ESA CCI Land cover dataset
485 (v2.0.7), Asia (mainly distributed in China, India, and Southeast Asia) ranked first among the five
486 continents which accounted for 34% of the global cropland area, followed by Europe (22%), Africa
487 (17%), North America (13%), South America (11%), and Oceania (3%). From a global perspective
488 (Figure 11b), it can be seen that a moderate meteorological drought may lead to approximately 44%
489 of the crop area falling under soil moisture drought, and the potential agricultural drought impacts
490 may be further enlarged (78% of the crop area for mild drought, and 10% for moderate drought)
491 given a severe meteorological drought (i.e., $\text{SPEI} \leq -2$). When the climatic condition becomes
492 extreme, the affected crop area may increase to 98%, with 60% for mild drought, 35% for moderate
493 drought, and 3% for severe drought. As the largest crop production base, Asia generally presents a
494 pattern similar to that of the globe conditioned on a moderate and severe meteorological drought.
495 However, for the extreme scenario, 7% of the crop area may suffer from severe drought, which is
496 two-fold higher than the global value (Figure 11c). With regard to drought duration, Figure 10
497 suggests the North China on average presents higher persistence of soil moisture droughts (may last
498 for three to six months) in response to meteorological droughts, and are more vulnerable to
499 agricultural disaster than other regions of Asia. For Europe, the affected crop area rapidly reaches up
500 to 97% under a moderate meteorological drought (i.e., $-1 < \text{SPEI} \leq -1.5$). Although such increments
501 are mostly under mild dry condition (Figure 11d), the moisture deficits may cause substantial

502 damages for crop production given the overall long extended duration (persist three to six months on
503 average, and negative anomalies of soil moisture over a long time especially in growing seasons may
504 be harmful for crop growths) of soil moisture droughts in this continent (Figure 10). In terms of
505 Africa and North America, the affected crop area conditioned on different values of SPEI1 is slightly
506 lower than that of the globe (Figures 11e, f), and the middle U.S. is more likely to suffer agricultural
507 losses in consideration of the longer duration of soil moisture droughts in this region (Figure 10).
508 The case for South America is not optimistic, either. Although Figure 10 suggests the soil moisture
509 drought for cropland in this continent would not persist for a long time, it may also induce large
510 agricultural impacts when referring to the affected area of each drought category. As shown in Figure
511 11g, 77% of the crop area may fall under drought given a mild meteorological drought (i.e.,
512 $-0.5 < \text{SPEI1} \leq -1$), while 69% may suffer from moderate drought when SPEI1 is no more than -2.
513 These proportion values are approximately two-fold higher than the global average, suggesting that
514 South America is more vulnerable to agricultural drought than other continents.

515

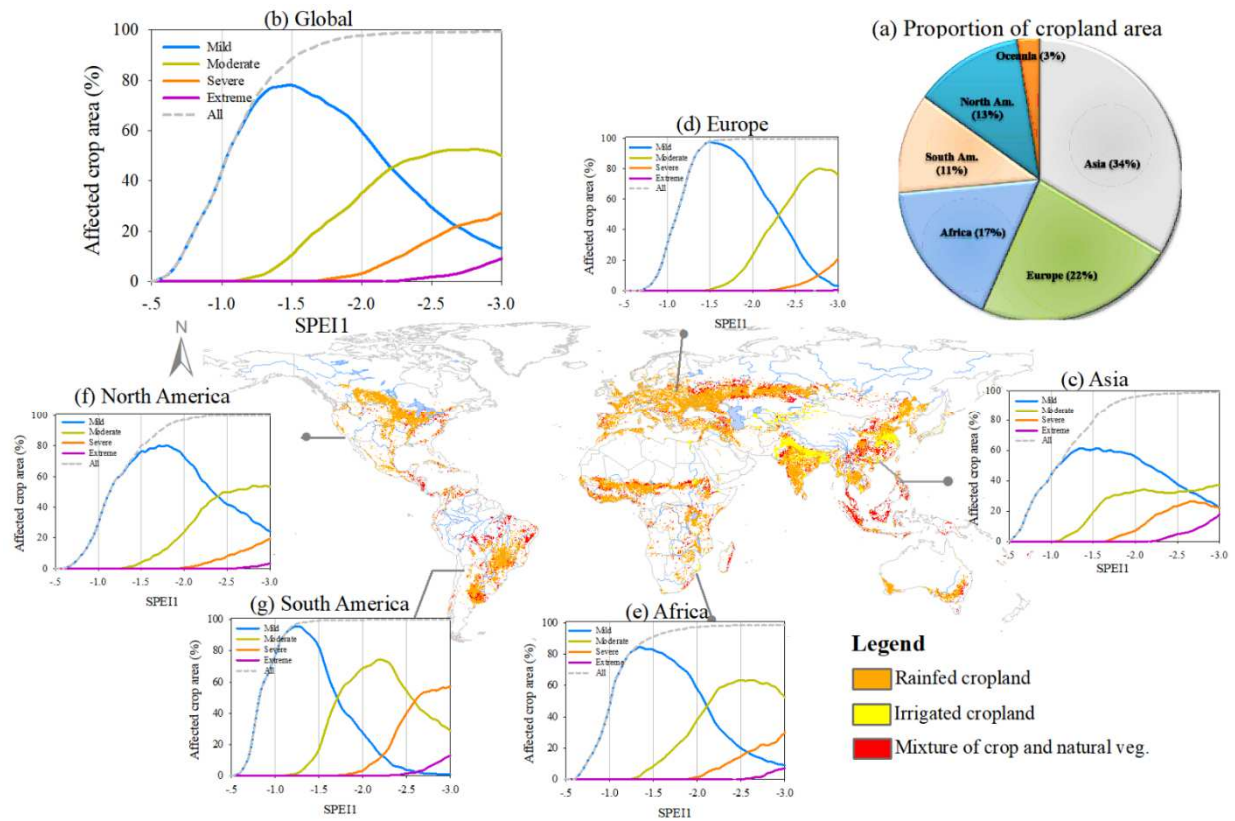


Figure 11. (a) Map of cropland distribution and proportion of cropland area for each continent. (b-g) The six rectangle panels show the proportion of affected crop area by mild, moderate, severe, and extreme drought given meteorological drought (indicated by SPEI1) for globe and each continent.

5 Discussion

5.1 Uncertainties

Some uncertainties may exist for the derived results regarding the selection of drought indices, time series of drought indices selected for constructing the conditional probability distribution, and GLDAS datasets. First, SPEI was employed in which precipitation and PET are treated equally for calculating the atmospheric moisture condition. Several previous studies claimed that the role of PET would be overestimated for estimating the dry status, particularly in arid zones (e.g., Cook et al., 2014; Ma et al., 2014). To evaluate the impact of SPEI on the probability of drought propagation, the

529 derived results were compared with those derived from SPI. We found in most regions of the world,
530 the probability of soil moisture drought derived from SPEI is very close to that derived from SPI,
531 which also suggest high probabilities in the Amazon rainforest, southern and eastern China,
532 Southeast Asia, southeastern U.S., West and Central Africa (see supplementary Figures S1 and S2).
533 Major differences are mainly located in water limited regions (e.g., the north Africa and northwestern
534 China), where the probability of soil moisture drought derived from SPEI is about 0.2 lower than
535 those from SPI (Figure 12). For arid areas, PET (i.e., the evaporative demand by the atmosphere
536 which represents the highest possible evapotranspiration) commonly far exceeds the actual moisture
537 supply, which may lead to the overestimation of the actual amount of water transferred to the
538 atmosphere (Raible et al., 2017). This leads to the climatic condition indicated by SPEI may present
539 a dry signal, but the soil moisture would not make a response, resulted in a lower probability value
540 than derived from SPI.

541 Second, the conditional probabilities were calculated by using complete time series of SPEI and
542 SMI, and there are uncertainties for estimating agricultural impacts (Figure 11) since the periods of
543 drought indices in growing seasons are more relevant for obtaining realistic implications of drought
544 propagation. To evaluate the influence of time series, the conditional probabilities by using data of
545 growing seasons were compared against those from complete time series. According to the frequency
546 distributions of the probability of soil moisture drought conditioned on moderate meteorological
547 drought (indicated by SPEI1 as an example) in Figure 13, it can be seen that the probability values
548 derived from data of growing seasons (red dashed line) overall increase by 0.1 comparing to those
549 derived from complete time series (the blue dashed line). Figure 14 exhibits the spatial distribution of
550 probability differences. Except for northern Canada, South Africa, western and southern Russia

551 where the absolute values of probability difference range between 0.1 and 0.2, for most regions of
552 the world, the differences of probability values would not exceed 0.05, plus or minus. This suggests
553 there would be tiny differences for estimated probability values between complete series and data of
554 growing seasons, but the associated global patterns of drought propagation (e.g., Figs. 4–11) would
555 not change in a significant way.

556 Finally, limitations of the land surface model (e.g., the inadequate consideration of snow-related
557 processes in cold regions and irrigation schemes) used in GLDAS also influence the quality of the
558 simulated datasets, and therefore for the conditional probability. For instance, comparing to in-situ
559 observations, Chen et al. (2013) found the four GLDAS models do not perform well in simulating
560 the surface soil moisture in central Tibetan Plateau. Agutu et al. (2017) assessed the performance of
561 GLDAS in characterizing agricultural drought in East African, and found GLDAS perform well in
562 Tanzania but mischaracterized the 2005-2006 drought in Kenyan and Ugandan. Zaitchik et al., (2010)
563 evaluated the performance of GLDAS in terms of global river discharge and found poor
564 performances in Congo River and high-latitude basins with snowmelt, which may have associated
565 effects on the accuracy of soil moisture estimates. In addition, the land surface models used in
566 GLDAS do not include the modules of irrigation schemes (a common drawback for most global
567 model simulation products), and there are uncertainties for irrigated cropland (Figure 11). For
568 instance, based on hydrological model simulations, several previous studies found irrigation leads to
569 increased evapotranspiration, decreased runoff, increased surface soil moisture, and decreased
570 streamflow (Haddeland et al., 2006; Ozdogan et al., 2010; Shah et al., 2021). From a climatic
571 perspective, several studies found irrigation would impact the water and energy balance near the
572 ground, making the near-surface atmosphere cooler and moister than non-irrigated areas (Kueppers

et al., 2007; Mishra et al., 2020). This cooling effect may create a favorable condition for drought propagation. Therefore, we may speculate that the probability of soil moisture drought under meteorological drought may be underestimated to some extent for irrigated croplands (e.g., northern India and the North China) in growing seasons. In spite of such limitations, Kim et al. (2020) found the soil moisture datasets from GLDAS generally perform better than certain satellite-based soil moisture products over moderately irrigated areas and is recommended for use when ground-based soil moisture data are not available. In future researches, it is necessary to conduct regional studies by using more sophisticated hydrological models in high resolution to reflect more realistic information of drought propagation for irrigated croplands.

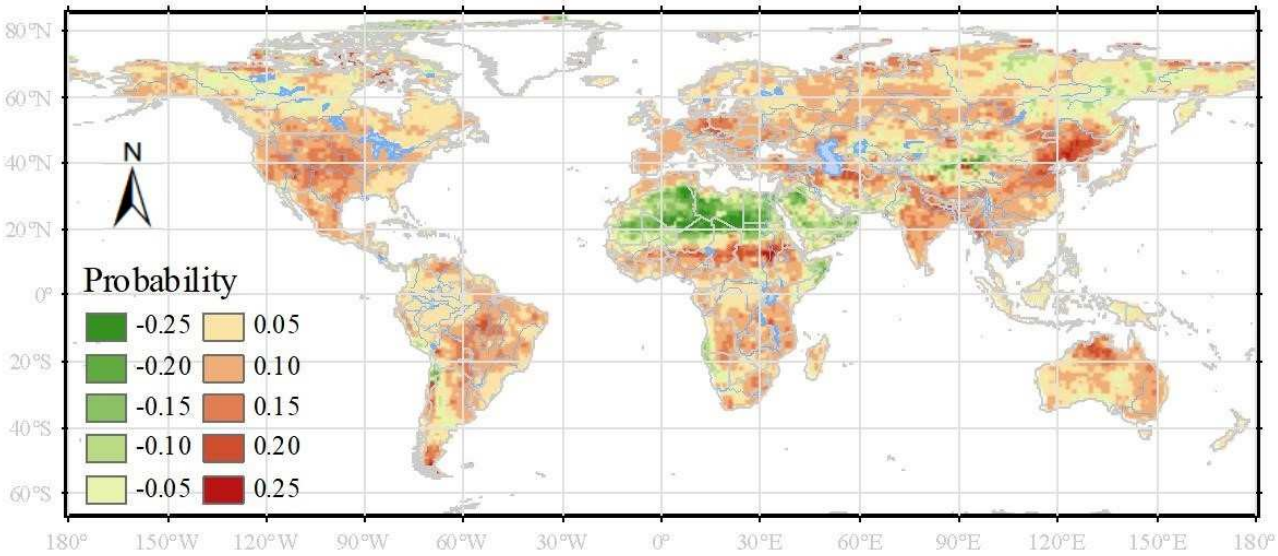


Figure 12. The difference of probability between SPEI1_SMI and SPI1_SMI. Positive values indicate the probability of soil moisture drought conditioned on SPEI1 is higher than that conditioned on SPI1, and vice versa.

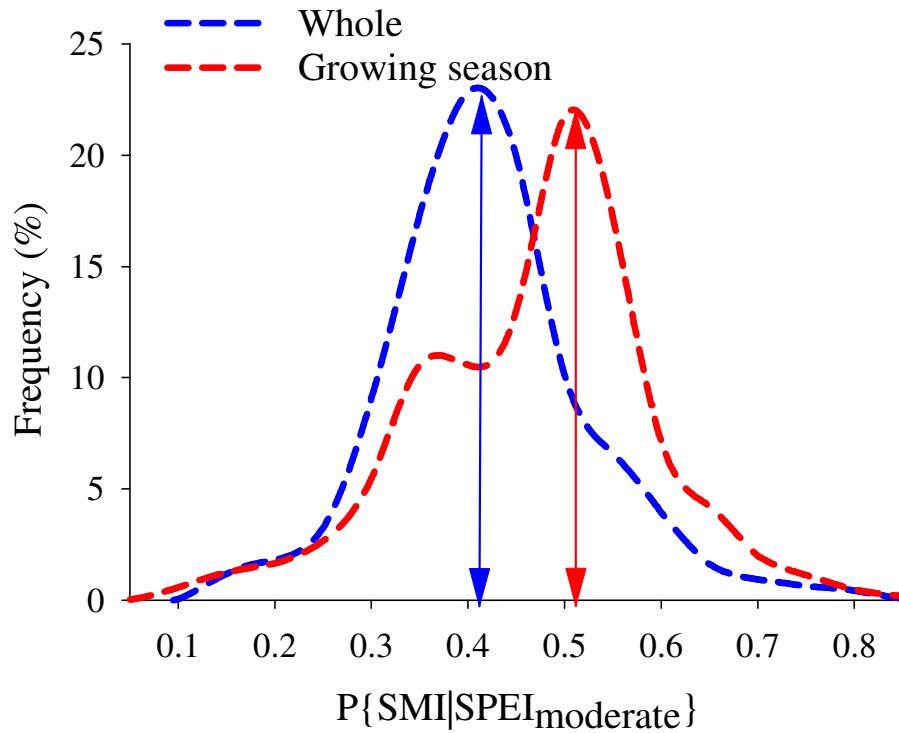


Figure 13. The frequency distribution of the probability of soil moisture drought conditioned on moderate meteorological drought (indicated by SPEI1) calculated from complete time series (blue dashed line) and time series of growing seasons (red dashed line).

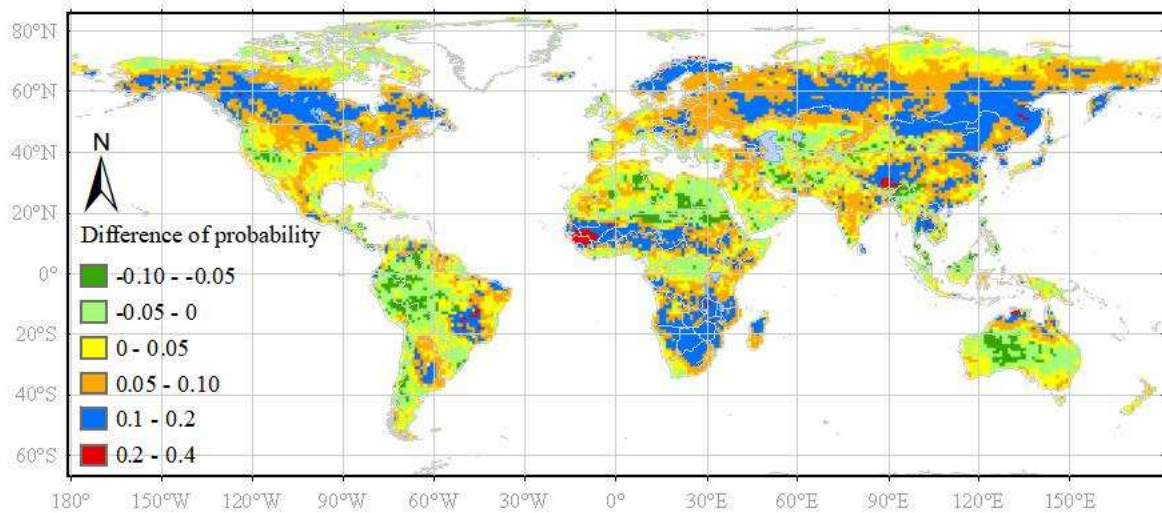


Figure 14. The difference of probability of soil moisture drought under moderate meteorological drought (indicated by SPEI1) by using the time series of growing seasons and complete time series.

5.2 Implications

Previous studies for analyzing the process of drought propagation from meteorological to agricultural, or hydrological mostly focused on the drought characteristics of different types, with some features associated with lagging, lengthening, and attenuation recognized (van Loon, 2015; Apurv et al. 2017; Liu et al. 2019). From a probabilistic perspective, this study uses a copula-based model to estimate the likelihood of soil moisture droughts conditioned on meteorological droughts of varying severity levels. This probability value could be interpreted as the strength of drought signal during the propagation process, where higher values imply that most of the drought information would be conveyed into the territorial hydrological system, and soil moisture droughts are more likely to be initiated if a meteorological drought occurs. It should be noted that this probability value has essential distinctions from aridity, or the frequency of droughts. As shown in Fig. 4, humid regions such as the Amazon rainforest and southern China present higher probability values than arid regions such as the north Africa. This phenomenon is not in contradiction with previous findings associated with the global patterns of drying trends (e.g., arid lands become more drier). Instead, Fig. 4 suggests that under the same meteorological drought conditions, regions in higher probability values like the Amazon rainforest and southern China are more vulnerable to soil moisture drought than north Africa.

In this study, we found the likelihood of soil moisture drought conditioned on SPEI was generally higher in Amazon rainforest, southern and eastern China, Southeast Asia, southeastern United States, and west Africa than in other regions. Comparison of SPEI and SMI in different climate regions suggests for high probability regions, the time series of SMI generally presents a good consistency with SPEI, and is sensitive to the changes of climate condition (Figures 6, 7). From

617 a climatic perspective, these regions are mostly in subtropical and tropical climate with sufficient
618 precipitation (i.e., water supply) and high temperature (i.e., evaporative demand). A climatic dry spell
619 suggests less water input (i.e., precipitation deficiency) is transported to the terrestrial system. At the
620 same time, PET may also increase (resulted from increased temperature, radiation, wind speed, etc.)
621 which can lead to boosted actual evapotranspiration with an extra loss of water from soil, vegetation,
622 or open water bodies (Seneviratne et al. 2010). All these contribute to the depletion of soil moisture,
623 which may further develop into a soil moisture drought. The moisture and heat condition produces a
624 close hydraulic relation among precipitation, evapotranspiration, and soil moisture (Chagas and
625 Chaffe, 2018; Yang et al. 2018), leading to a strong nexus between meteorological and soil moisture
626 droughts. Similar to previous studies (e.g., Vicente-Serrano et al., 2005; Peña-Gallardo et al., 2019)
627 which related accumulated SPEI of varying time scales to hydrological variables, the time scale
628 effect of droughts was also observed in this study. As shown in Figure 5, the North America and
629 Europe present significant increments in the probability values of soil moisture drought conditioned
630 on SPEI3 than those conditioned on SPEI1 (Figure 4). This reflects the cumulative effects of
631 meteorological water deficits on drought propagation. Meanwhile, the enhanced probabilistic link
632 under SPEI3 in these regions also indicate a lagged (or slower) response of soil moisture to the
633 climatic condition than regions of high probability values (e.g., the Amazon rainforest). This
634 phenomenon agrees with findings in Bachmair et al. (2018) which also suggest the 3-month
635 meteorological drought indices are best linked to agricultural and forest droughts.

636 Apart from the current meteorological condition, we found that the antecedent soil moisture
637 status also influences the probability of drought propagation. Comparing to the patterns in Fig. 4, Fig.
638 8 shows that under the joint condition of meteorological droughts (indicated by SPEI1) and normal

639 antecedent soil moisture status ($-0.5 < \text{SMI}_{\text{lag-1}} < 0.5$), the likelihood of soil moisture drought
640 significantly decreases in the Northern China, the U.S. Midwest, Australia, Russia, and Canada.
641 According to the geographical definition, these regions are mostly “transitional regions”, where soil
642 moisture has been shown to have a strong influence on the land-atmospheric interactions via the soil
643 moisture-temperature coupling effect (Seneviratne et al., 2010). This includes the feedback loops that
644 a positive anomaly of temperature in response to a negative soil moisture anomaly, and increased
645 temperature contributes to a higher vapor pressure deficit and evaporative demand, which possibly
646 leading to a further decrease in soil moisture (van Loon, 2015). This shows the key role of soil
647 moisture in affecting the land energy and water balances in transitional regions, which is also the
648 core content of drought propagation process from meteorological to soil moisture.

649 Findings in this study highlight the varied responses of soil moisture to meteorological drought
650 under different moisture conditions (both the atmospheric and the antecedent soil moisture status).
651 Despite the uncertainties discussed above, the probability of drought propagation provides some
652 implications for recognizing the potential impacts of soil droughts (Figure 11), which also has
653 scientific guidance for water security assessment and drought management. Ongoing analysis on the
654 propagation behavior in different stages (e.g., onset, recover) of drought, and the seasonality in
655 different regions are needed for improving the understanding of drought propagation mechanism.

656 **6 Conclusions**

657 In this study, a copula based probabilistic model (i.e., the meta-Gaussian and vine copulas for
658 bivariate and trivariate cases, respectively) was employed to illustrate the propagation of
659 meteorological to soil moisture drought. Three different cases, including the probability of soil
660 moisture drought (SMI) conditioned on 1-month meteorological drought (SPEI1), seasonal

661 meteorological drought (SPEI3), and the joint effects of SPEI1 and antecedent soil moisture
662 conditions (SMI_{lag-1}) were investigated. Based on the established conditional joint distributions, the
663 characteristics of drought propagation, including the probable magnitude of soil moisture drought
664 (i.e., the extent to which the soil moisture condition may be drying), the extended duration of soil
665 moisture droughts in response to meteorological droughts, and potential agricultural drought risks
666 were analyzed.

667 Results showed that the likelihood of soil moisture drought given SPEI1 tended to be higher
668 when the meteorological condition was drier. Spatially, the Amazon rainforest, southern and eastern
669 China, Southeast Asia, southeastern United States, and west Africa were more sensitive to
670 meteorological drought with high probabilities. Under the impact of seasonal meteorological drought,
671 the probability of soil moisture drought occurrence overall increased with significant increments in
672 Europe and North America. In contrast, the Qinghai-Tibet Plateau in China, northern Russia, and
673 North Africa were less affected and maintained rather low probability values when conditioned on
674 SPEI1. The antecedent soil moisture condition also influenced the probability of drought propagation,
675 and this effect was particularly significant for Northern China, Russia, the American Midwest,
676 Canada, and Australia. In terms of the probable magnitude of soil moisture drought, a moderate
677 meteorological drought may cause little effect on the soil moisture condition, while severe and
678 extreme meteorological drought may lead to mild soil moisture drought for most parts of the world.
679 Investigation on drought duration suggests the persistence of drought would be lengthened during the
680 drought propagation process from meteorological to soil moisture, with spatially longer duration of
681 soil moisture drought in the western U.S., northern Canada, South Africa, eastern Europe, North
682 China, northern Russia, and most parts of Australia.

683 A further investigation into the affected cropland area revealed that a moderate meteorological
684 drought may lead to approximately 44% of the cropland falling under drought, and the potential
685 agricultural drought impacts may be further enlarged (78% of the crop area for mild drought, and
686 10% for moderate drought) given severe meteorological drought. When the climatic condition
687 becomes extreme, the affected cropland area may increase to 98%, with 60% for mild drought, 35%
688 for moderate drought, and 3% for severe drought. With overall higher proportions of the affected
689 cropland area than the global averages, South America is most likely to suffer from agricultural
690 drought under meteorological drought, followed by Asia, Africa and North America. Europe is more
691 sensitive to meteorological drought with 97% of the crop area may be affected under a moderate
692 meteorological drought. Although these are mostly under mild dry condition, they could be
693 damaging to agricultural production given the overall long duration of soil moisture droughts.

694 From a global perspective, the above results provide a spatial image of the likelihood of soil
695 moisture drought conditioned on moderate, severe, and meteorological droughts, which also have
696 implications for drought prevention and mitigation strategies in the context of global warming.

697 **CRedit author statement**

698 Ye Zhu: Conceptualization, Validation, Investigation, Writing-Original draft. Yi Liu: Data curation,
699 Methodology, Software, Resources. Wen Wang: Formal analysis, Supervision. Vijay P. Singh:
700 Writing- Reviewing and Editing. Liliang Ren: Software, Resources.

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